

MOST IMPORTANT TOPICS FOR GATE DA

Subjectwise Analysis

ANJALI CHAUHAN

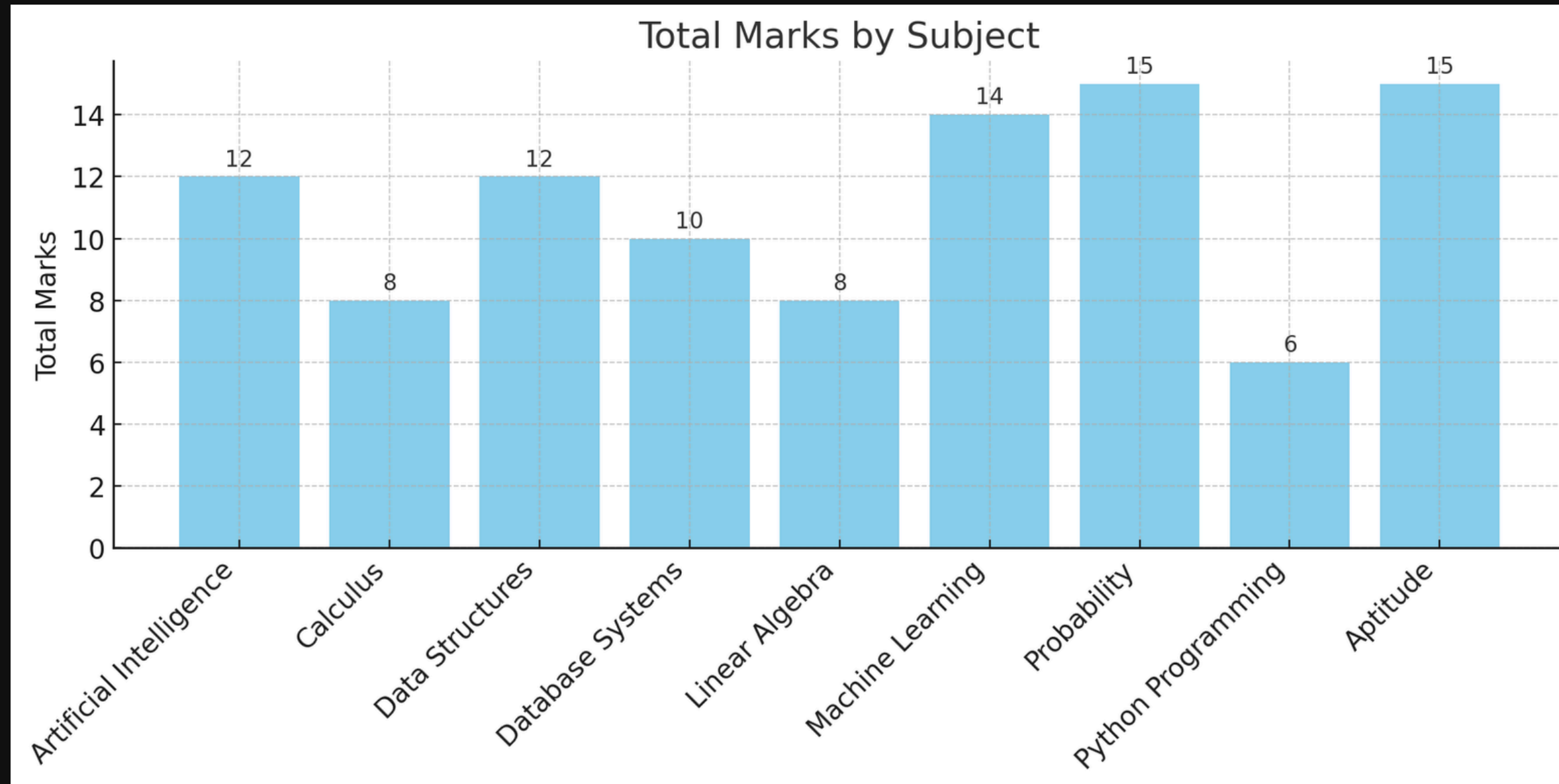
GATE AIR 13

GATE DA 2024 question paper and sample paper analysis

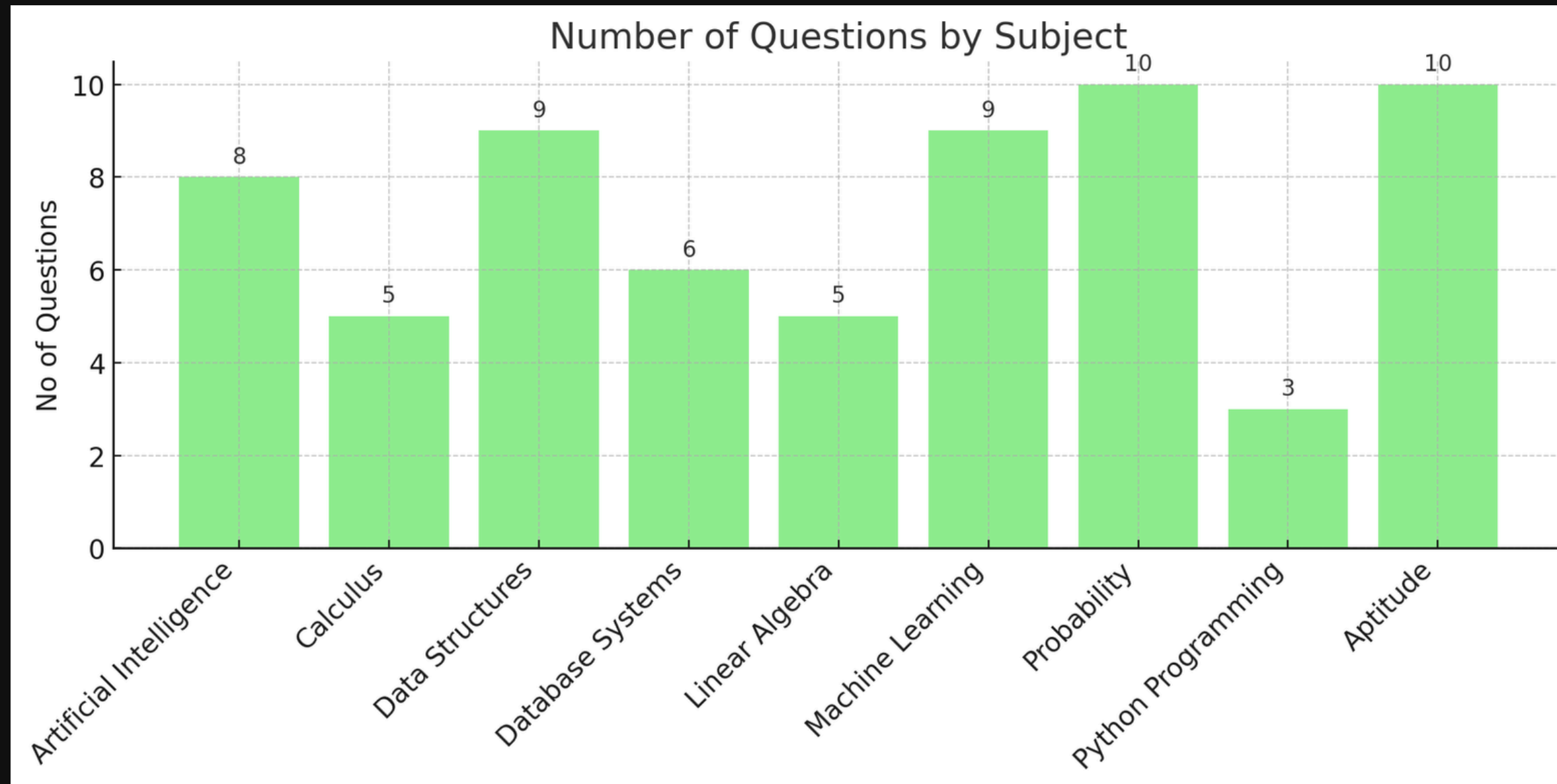
<https://gate2024.iisc.ac.in/wp-content/uploads/2024/DA24S1.pdf>

<https://gate2024.iisc.ac.in/wp-content/uploads/2023/10/DataScienceAISampleQuestionPaper.pdf>

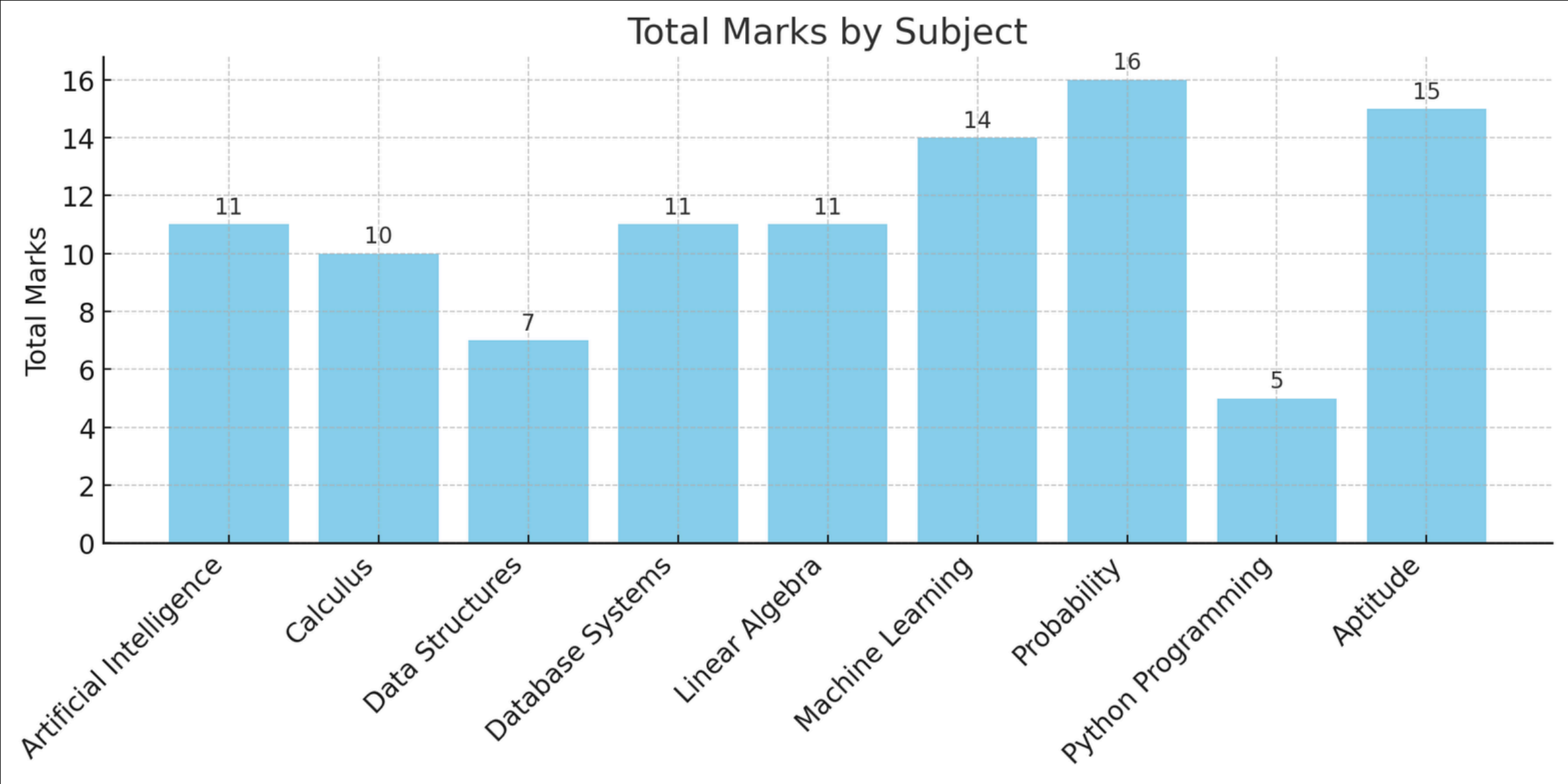
GATE DA 2024 Question Paper



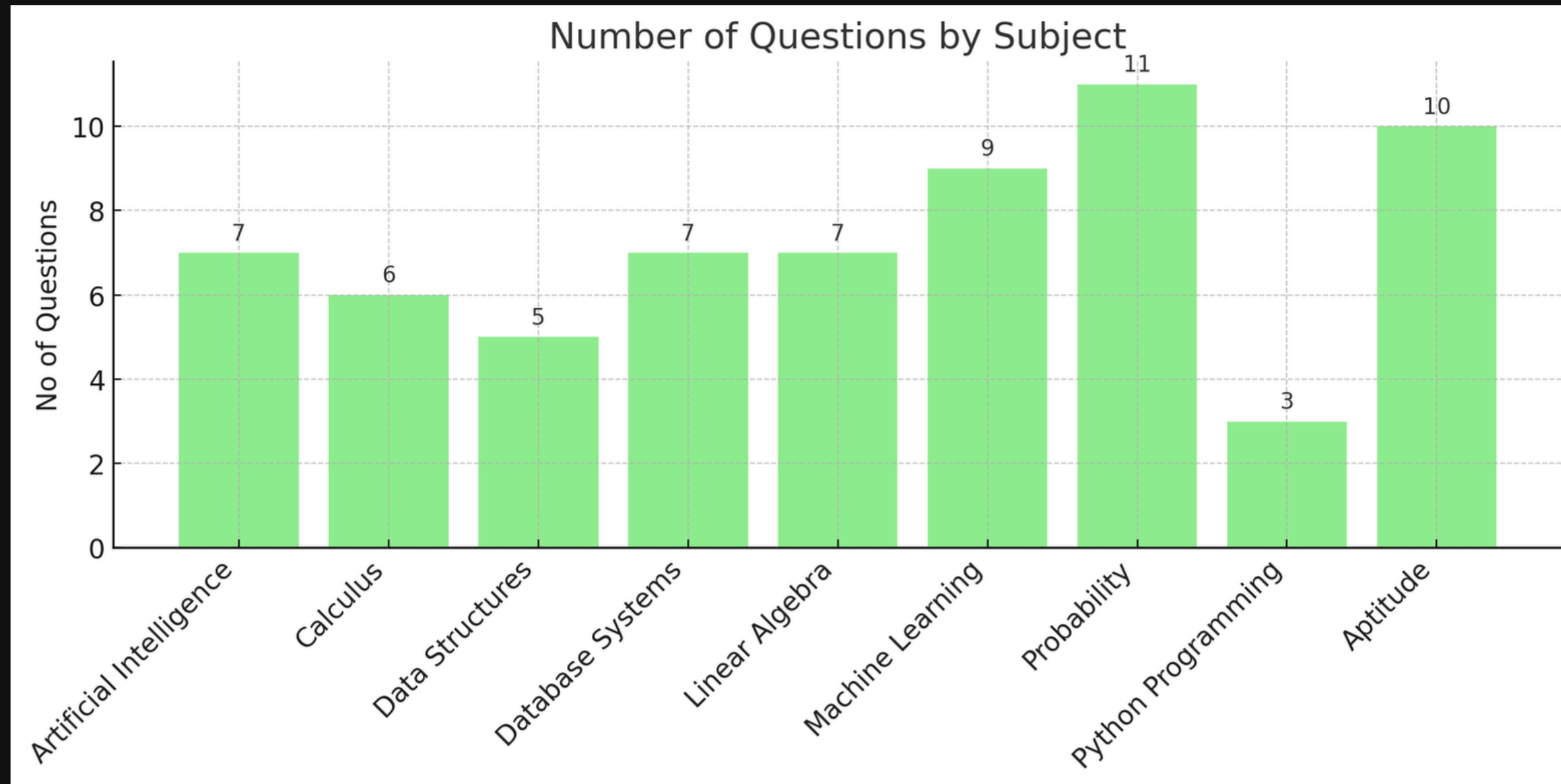
GATE DA 2024 Question Paper



GATE DA 2024 Sample Paper



GATE DA 2024 Sample Paper



NOTE

All subjects are equally important!

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This was just 1 year data & we don't have enough
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This was just 1 year data & we don't have enough
data to generalize anything
(Machine Learning doesn't allow us).

Probability and Statistics

Probability and Statistics: Counting (permutation and combinations), probability axioms, Sample space, events, independent events, mutually exclusive events, marginal, conditional and joint probability, Bayes Theorem, conditional expectation and variance, mean, median, mode and standard deviation, correlation, and covariance, random variables, discrete random variables and probability mass functions, uniform, Bernoulli, binomial distribution, Continuous random variables and probability distribution function, uniform, exponential, Poisson, normal, standard normal, t-distribution, chi-squared distributions, cumulative distribution function, Conditional PDF, Central limit theorem, confidence interval, z-test, t-test, chi-squared test.

Topics asked in GATE DA 2024

Probability and Statistics

- Random Variables
- Distributions (Poisson, Normal, Exponential)
- Conditional Probability
- Covariance

Important Topics

Probability and Statistics

- Random Variables
- Distributions (Poisson, Normal, Exponential)
- Conditional Probability (Bayes Theorem)
- Joint Distributions

Linear Algebra

Linear Algebra: Vector space, subspaces, linear dependence and independence of vectors, matrices, projection matrix, orthogonal matrix, idempotent matrix, partition matrix and their properties, quadratic forms, systems of linear equations and solutions; Gaussian elimination, eigenvalues and eigenvectors, determinant, rank, nullity, projections, LU decomposition, singular value decomposition.

Topics asked in GATE DA 2024

Linear Algebra

- Eigenvalues and Eigenvectors
- Subspaces
- Matrix Algebra

Important Topics

Linear Algebra

- Eigenvalues/Eigenvectors
- Subspaces
- Determinants
- System of linear equations

Calculus and Optimization

Calculus and Optimization: Functions of a single variable, limit, continuity and differentiability, Taylor series, maxima and minima, optimization involving a single variable.

Topics asked in GATE DA 2024

Calculus and Optimization

- Differentiation and Minima/Maxima
- Limits

Important Topics

Calculus and Optimization

- Limits
- Continuity and Differentiability
- Maxima and minima
- Integration

Programming, Data Structures and Algorithms

Programming, Data Structures and Algorithms: Programming in Python, basic data structures: stacks, queues, linked lists, trees, hash tables; Search algorithms: linear search and binary search, basic sorting algorithms: selection sort, bubble sort and insertion sort; divide and conquer: mergesort, quicksort; introduction to graph theory; basic graph algorithms: traversals and shortest path.

Topics asked in GATE DA 2024

Programming, Data Structures and Algorithms

- Depth-First Search (DFS)
- Sorting Algorithms (Quicksort)
- Binary Search
- Hashing
- Python Code (Recursion)
- Topological Sorting in DAG

Important Topics

Programming, Data Structures and Algorithms

- Recursion
- Stacks, Queues
- Sorting Algorithms
- DFS, BFS, shortest path
- Tree Properties: Binary trees, BST
- Hashing

Database Management and Warehousing

Database Management and Warehousing: ER-model, relational model: relational algebra, tuple calculus, SQL, integrity constraints, normal form, file organization, indexing, data types, data transformation such as normalization, discretization, sampling, compression; data warehouse modelling: schema for multidimensional data models, concept hierarchies, measures: categorization and computations.

Topics asked in GATE DA 2024

Database Management and Warehousing

- Relational Algebra
- SQL Queries
- Indexing

Important Topics

Database Management and Warehousing

- SQL Queries
- Normalization
- Indexing
- Relational Algebra
- ER Model

Machine Learning

Database Management and Warehousing: ER-model, relational model: relational algebra, tuple calculus, SQL, integrity constraints, normal form, file organization, indexing, data types, data transformation such as normalization, discretization, sampling, compression; data warehouse modelling: schema for multidimensional data models, concept hierarchies, measures: categorization and computations.

Topics asked in GATE DA 2024

Machine Learning

- Support Vector Machines (SVM)
- Naive Bayes Classification
- Logistic Regression
- K means Clustering
- Linear Discriminant Analysis

Important Topics

Machine Learning

- Linear Regression
- Logistic Regression
- Support Vector Machines (SVM)
- K means, KNN
- Naïve Bayes

Artificial Intelligence

AI: Search: informed, uninformed, adversarial; logic, propositional, predicate; reasoning under uncertainty topics — conditional independence representation, exact inference through variable elimination, and approximate inference through sampling.

Topics asked in GATE DA 2024

Artificial Intelligence

- A* Search
- Alpha-Beta Pruning
- Bayesian Networks
- Propositional Logic

Important Topics

Artificial Intelligence

- Logic
- Bayesian network Representation
- A* Algorithm

Aptitude

15 marks every year

Aptitude

- Solve Last 15 year PYQs of all the branches.

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To All GATE DA Aspirants,

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- Feel happy, because syllabus is less.

To All GATE DA Aspirants,

- Feel happy, because syllabus is less.
- ML is not the only subject in GATE DA.
So if you don't know ML, you can skip it.

THANK YOU

**Please let me know in the comments if you have any doubts related to GATE preparation.
Would be more than happy to answer them :)**